

Quantum Pinsker Inequality, the Reducing Kraus Map, and References

Terminology. The inequality

$$D(X\|Y) \geq \frac{1}{2} \|X - Y\|_1^2$$

for trace-one positive semidefinite matrices is usually called the *quantum Pinsker inequality*. One also sees the names *noncommutative Pinsker inequality* or *Pinsker inequality for quantum relative entropy*, but *quantum Pinsker inequality* is the standard name.

Convention. All logarithms are natural. With this convention the constant is $1/2$. If one uses base-2 logarithms, the constant becomes $1/(2 \ln 2)$.

1 Classical Pinsker

For probability vectors $p = (p_i)_{i=1}^m$ and $q = (q_i)_{i=1}^m$,

$$D(p\|q) := \sum_{i=1}^m p_i \log \frac{p_i}{q_i}.$$

Then

$$D(p\|q) \geq \frac{1}{2} \|p - q\|_1^2. \quad (1)$$

For arbitrary nonnegative vectors $x, y \in \mathbb{R}_+^m$, the appropriate unnormalized divergence is

$$I(x\|y) := \sum_{i=1}^m \left(x_i \log \frac{x_i}{y_i} - x_i + y_i \right).$$

If

$$a := \|x\|_1, \quad b := \|y\|_1, \quad p := \frac{x}{a}, \quad q := \frac{y}{b},$$

then

$$I(x\|y) = a D(p\|q) + \left(a \log \frac{a}{b} - a + b \right), \quad (2)$$

hence

$$I(x\|y) \geq \frac{1}{2a} \left\| x - \frac{a}{b} y \right\|_1^2 + \left(a \log \frac{a}{b} - a + b \right). \quad (3)$$

2 Matrix relative entropy

Let $X, Y \in \mathbb{C}^{n \times n}$ be positive semidefinite. Define

$$D(X\|Y) := \begin{cases} \operatorname{Tr}[X(\log X - \log Y) - X + Y], & \operatorname{supp}(X) \subseteq \operatorname{supp}(Y), \\ +\infty, & \text{otherwise.} \end{cases}$$

If $\text{Tr } X = \text{Tr } Y = 1$, this reduces to

$$D(X\|Y) = \text{Tr}[X(\log X - \log Y)].$$

The norm on matrices is the trace norm

$$\|A\|_1 := \text{Tr } |A|.$$

3 Quantum Pinsker inequality

Theorem 1 (Quantum Pinsker inequality). *If $X, Y \geq 0$ and $\text{Tr } X = \text{Tr } Y = 1$, then*

$$D(X\|Y) \geq \frac{1}{2} \|X - Y\|_1^2. \quad (4)$$

Remark 1. With logarithms in base 2, the same statement becomes

$$D_2(X\|Y) \geq \frac{1}{2 \ln 2} \|X - Y\|_1^2.$$

4 The reducing two-point map

The clean way to prove Theorem 1 is to reduce it to the classical two-point case.

4.1 Positive and negative parts

Let

$$\Delta := X - Y.$$

Since Δ is Hermitian, there is an orthonormal basis of eigenvectors and real eigenvalues:

$$\Delta = \sum_{j=1}^n \lambda_j u_j u_j^*.$$

Define

$$\Delta_+ := \sum_{\lambda_j > 0} \lambda_j u_j u_j^*, \quad \Delta_- := \sum_{\lambda_j < 0} (-\lambda_j) u_j u_j^*.$$

Then

$$\Delta = \Delta_+ - \Delta_-, \quad |\Delta| = \Delta_+ + \Delta_-.$$

Let

$$P := \sum_{\lambda_j > 0} u_j u_j^*,$$

the orthogonal projector onto the positive spectral subspace of Δ .

4.2 The map to a two-point distribution

Lemma 1 (Reducing Kraus map). *Let $X, Y \geq 0$ with $\text{Tr } X = \text{Tr } Y = 1$, and let P be the orthogonal projector onto the positive spectral subspace of $X - Y$. Define*

$$\phi(A) := (\text{Tr}(PA), \text{Tr}((1 - P)A)), \quad A \in \mathbb{C}^{n \times n}. \quad (5)$$

Define also the matrix-valued map

$$\Phi(A) := \text{Tr}(PA) e_1 e_1^\top + \text{Tr}((1 - P)A) e_2 e_2^\top, \quad (6)$$

where e_1, e_2 are the standard basis vectors of $\mathbb{C}^2$.

Then:

- (i) ϕ is linear;
- (ii) if $A \geq 0$, then $\phi(A) \in \mathbb{R}_+^2$;
- (iii) if $\text{Tr } A = 1$, then $\phi(A)$ is a probability vector;
- (iv) with

$$p := \phi(X), \quad q := \phi(Y),$$

one has

$$\|p - q\|_1 = \|X - Y\|_1; \quad (7)$$

- (v) Φ is completely positive and trace-preserving;
- (vi) Φ has a Kraus representation.

Proof. Linearity of ϕ is immediate.

If $A \geq 0$, then $P \geq 0$ and $\mathbf{1} - P \geq 0$, so

$$\text{Tr}(PA) \geq 0, \quad \text{Tr}((\mathbf{1} - P)A) \geq 0.$$

Hence $\phi(A) \in \mathbb{R}_+^2$.

If $\text{Tr } A = 1$, then

$$\text{Tr}(PA) + \text{Tr}((\mathbf{1} - P)A) = \text{Tr}(A) = 1,$$

so $\phi(A)$ is a probability vector.

Now

$$p - q = (\text{Tr}(P\Delta), \text{Tr}((\mathbf{1} - P)\Delta)), \quad \Delta = X - Y.$$

Because P projects onto the positive spectral subspace of Δ ,

$$\text{Tr}(P\Delta) = \text{Tr}(\Delta_+), \quad \text{Tr}((\mathbf{1} - P)\Delta) = -\text{Tr}(\Delta_-).$$

Therefore

$$\begin{aligned} \|p - q\|_1 &= |\text{Tr}(P\Delta)| + |\text{Tr}((\mathbf{1} - P)\Delta)| \\ &= \text{Tr}(\Delta_+) + \text{Tr}(\Delta_-) \\ &= \text{Tr} |\Delta| \\ &= \|\Delta\|_1 = \|X - Y\|_1. \end{aligned}$$

This proves (6).

For the Kraus form, set

$$K_1 := e_1 P, \quad K_2 := e_2 (\mathbf{1} - P).$$

Then

$$K_1^* K_1 = P, \quad K_2^* K_2 = \mathbf{1} - P, \quad K_1^* K_1 + K_2^* K_2 = \mathbf{1}.$$

Moreover, for every A ,

$$K_1 A K_1^* = \text{Tr}(PA) e_1 e_1^\top, \quad K_2 A K_2^* = \text{Tr}((\mathbf{1} - P)A) e_2 e_2^\top.$$

Hence

$$\Phi(A) = K_1 A K_1^* + K_2 A K_2^*.$$

So Φ is given by a Kraus representation, therefore it is completely positive; and the identity

$$K_1^* K_1 + K_2^* K_2 = \mathbf{1}$$

shows that it is trace-preserving. □

Remark 2. The vector-valued map ϕ is the diagonal of the matrix-valued map Φ . Complete positivity is naturally stated for Φ , because its codomain is a matrix algebra. One then identifies the diagonal 2×2 matrix $\Phi(A)$ with its diagonal vector $\phi(A)$.

5 Proof of the quantum Pinsker inequality

The key input is monotonicity of matrix relative entropy under completely positive trace-preserving maps:

$$D(X\|Y) \geq D(\Phi(X)\|\Phi(Y)).$$

By Lemma 1, the matrices $\Phi(X)$ and $\Phi(Y)$ are diagonal with diagonals

$$p = \phi(X), \quad q = \phi(Y).$$

Therefore

$$D(\Phi(X)\|\Phi(Y)) = D(p\|q).$$

Applying the classical Pinsker inequality (1), we obtain

$$D(X\|Y) \geq D(p\|q) \geq \frac{1}{2} \|p - q\|_1^2.$$

Using again Lemma 1,

$$\|p - q\|_1 = \|X - Y\|_1.$$

Hence

$$D(X\|Y) \geq \frac{1}{2} \|X - Y\|_1^2.$$

This proves Theorem 1.

6 Extension to arbitrary positive semidefinite matrices

Now let $A, B \geq 0$ be arbitrary, and set

$$a := \operatorname{Tr} A, \quad b := \operatorname{Tr} B.$$

Assume $a, b > 0$ and define

$$X := \frac{A}{a}, \quad Y := \frac{B}{b}.$$

Then

$$D(A\|B) = a D(X\|Y) + \left(a \log \frac{a}{b} - a + b \right). \quad (8)$$

Combining this with Theorem 1 yields

$$D(A\|B) \geq \frac{1}{2a} \left\| A - \frac{a}{b} B \right\|_1^2 + \left(a \log \frac{a}{b} - a + b \right). \quad (9)$$

If $\operatorname{Tr} A = \operatorname{Tr} B = a$, this simplifies to

$$D(A\|B) \geq \frac{1}{2a} \|A - B\|_1^2. \quad (10)$$

7 Historical and bibliographic remarks

Remark 3. Historically, the inequality (4) is commonly attributed to Hiai, Ohya, and Tsukada, who stated it in 1981 in their paper *Sufficiency, KMS condition and relative entropy in von Neumann algebras*. A convenient modern source for this attribution is Watrous's textbook, which explicitly says that the quantum Pinsker inequality appears in that paper.

Remark 4. Watrous also notes that the same inequality can be obtained as a special case of a more general theorem due to Uhlmann from 1977. So there are two natural historical references:

- (1) the direct 1981 reference of Hiai–Ohya–Tsukada;
- (2) the earlier 1977 paper of Uhlmann containing a more general result from which quantum Pinsker follows.

Remark 5. For modern references, a very convenient textbook source is:

- (1) J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018.

For broader background on quantum entropy and relative entropy, one may also consult:

- (1) M. Ohya and D. Petz, *Quantum Entropy and Its Use*, Springer, 1993.

For a modern set of lecture notes that states the inequality explicitly, see:

- (1) M. Wirth, *Trace Inequalities and Quantum Entropy*, 2025 lecture notes.

8 References

References

- [1] F. Hiai, M. Ohya, and M. Tsukada, *Sufficiency, KMS condition and relative entropy in von Neumann algebras*, Pacific Journal of Mathematics **96** (1981), no. 1, 99–109.
- [2] A. Uhlmann, *Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity in an interpolation theory*, Communications in Mathematical Physics **54** (1977), no. 1, 21–32.
- [3] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018.
- [4] M. Ohya and D. Petz, *Quantum Entropy and Its Use*, Springer, 1993.
- [5] M. Wirth, *Trace Inequalities and Quantum Entropy*, lecture notes, 2025.